Extended reading — Week 02: Bellman Equations and Dynamic Programming

November 8, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Bellman (1957): Dynamic Programming

Primary source or book record: https://press.princeton.edu/books/paperback/9780691146683/dynamic-programming.

What to learn. The principle of optimality decomposes a decision sequence into its immediate consequence and an optimal continuation. This is the conceptual starting point for Bellman backups.

What the result requires. The state must retain what is needed to predict rewards and transitions. A history-dependent problem may need an augmented state before the recursion is valid. Finite-horizon values also depend on remaining time.

Notebook investigation. Use `exercise_1_bellman_operator`. With one action, reward 2, successor value 3 and discount 0.9, a backup gives 4.7. Identify which quantity is an input estimate and which is an environment property.

2. Howard (1960): Dynamic Programming and Markov Processes

Primary source or book record: https://books.google.com/books?id=fXJEAAAAIAAJ.

What to learn. Policy iteration alternates evaluation of a fixed policy and greedy improvement. Reading the two stages separately prevents confusion between evaluating a policy and solving the control problem.

What the result requires. For a finite discounted MDP, exact evaluation and consistent tie handling support termination. There is no general guarantee that policy iteration requires only logarithmically many improvement steps. This catalog link identifies the book; it is not a page-level proof source.

Notebook investigation. Compare `exercise_3_policy_iteration` with `exercise_6_policy_evaluation_closed_form`. Check $(I - \gamma P_\pi)V = r_\pi$ before interpreting a policy switch.

3. Puterman (1994): Markov Decision Processes

Primary source or book record: https://onlinelibrary.wiley.com/doi/book/10.1002/9780470316887.

What to learn. Provides a systematic treatment of finite-state decision processes. Use the discounted chapters to follow contraction, existence of optimal stationary policies and evaluation/improvement arguments.

What the result requires. Average-reward, undiscounted episodic and discounted results have different hypotheses. A theorem for finite discounted models cannot be transferred to arbitrary neural approximation by changing the symbols.

Notebook investigation. Derive $\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty$. Use `exercise_5_contraction_factor` to check an upper bound, not demand equality at each sweep.

4. Tsitsiklis and Van Roy (1997): An Analysis of Temporal-Difference Learning with Function Approximation

Primary source or book record: https://web.mit.edu/~jnt/www/Papers/J063-97-bvr-td.pdf.

What to learn. Analyses linear TD under specified on-policy sampling and stochastic-approximation conditions. The limit solves a projected Bellman equation involving a stationary-distribution-weighted L_2 projection.

What the result requires. The TD fixed point is generally not the orthogonal projection of the true value function. The relevant projection argument is not a weighted sup-norm contraction. Off-policy sampling and nonlinear approximation need separate analysis.

Notebook investigation. For feature matrix Φ, distinguish ΠV^π from the solution to $v = \Pi T^\pi v$. Explain why changing the sampling weights changes the projection geometry.

5. Munos and Szepesvari (2008): Finite-Time Bounds for Fitted Value Iteration

Primary source or book record: https://jmlr.org/papers/v9/munos08a.html.

What to learn. Studies sampling-based fitted value iteration with a generative model, separating statistical approximation and error propagation. The analysis includes inherent Bellman error and concentrability of future state distributions.

What the result requires. A small regression error on one sampling distribution does not automatically imply small control error under a different occupancy. The bounds are not a plug-in convergence theorem for the local deep-Q implementation.

Notebook investigation. Sketch a distribution that samples one state frequently and misses a valuable reachable state. Explain why more samples from the same distribution may leave the relevant approximation failure unresolved.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.

Statistics reading bridge

A Bellman expectation need never occur on a trajectory. Read conditional expectation before expanding the Bellman sum. Label which variables are averaged and which estimates are held fixed. Use the discrete-time chain section for transition-law meaning, while retaining the RL sources for control and contraction results.

Checked textbook locations: Siegrist § 4.7, PDF p.507; § 16.3, PDF p.1322. https://stats.libretexts.org/@go/page/10163. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.